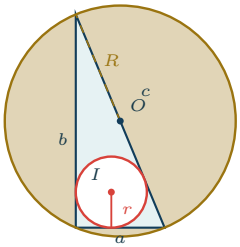
GEOMETRY ON ONE PAGE

# The annulus between circumcircle and incircle

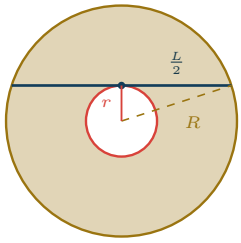
Its area is $\pi(R^2 - r^2)$, a multiple of $\frac{\pi}{4}$ for every triple, and two triples can share it.

## 1 / THE RULE AND THE PICTURE

### The region between the two circles has area $\pi(R^2 - r^2) = \dfrac{\pi}{4}(c^2 - 4r^2)$




$O$ circumcentre, $I$ incentre




$R^2 = r^2 + (\frac{L}{2})^2$

**The letters.** $a, b$ are the legs and $c$ the hypotenuse. The hypotenuse is a diameter of the circumcircle, so its radius is $R = \frac{c}{2}$, and the incircle has radius $r = \frac{a+b-c}{2}$. Then

$$A = \pi(R^2 - r^2) = \tfrac{\pi}{4}(2c - a - b)(a + b)$$

For $5, 12, 13$ this is $\pi(42.25 - 4) = \frac{153}{4}\pi \approx 120.17$. The centres $O$ and $I$ are $4.03$ apart, yet only the radii matter, so the concentric annulus on the right has the same area: its chord $L$ touching the inner circle gives $R^2 = r^2 + (\frac{L}{2})^2$, so $A = \pi(\frac{L}{2})^2$.

## 2 / THE GENERAL EQUATION

### Why a factor $k$ appears, and what $u$ and $v$ are

Euclid's formula $(m^2 - n^2,\ 2mn,\ m^2 + n^2)$, with $m > n$ coprime and of opposite parity, produces each *primitive* triple (no common factor) exactly once, such as $3, 4, 5$ and $5, 12, 13$. Every other whole-number right triangle is a primitive one enlarged by a whole number $k$, as $6, 8, 10 = 2 \cdot (3, 4, 5)$ and $15, 36, 39 = 3 \cdot (5, 12, 13)$, so $k(m^2 - n^2,\ 2mn,\ m^2 + n^2)$ lists all of them; $m$ and $n$ are the two numbers fed into the formula, not part of the triangle. Enlarging by $k$ multiplies every length by $k$ and the annulus area by $k^2$.

| triple | $k$ | $(m, n)$ | $R$ | $r$ | $N = 4A/\pi$ | $A/\pi$ |
|---|---|---|---|---|---|---|
| $3, 4, 5$ | 1 | $(2, 1)$ | 2.5 | 1 | 21 | 21/4 |
| $5, 12, 13$ | 1 | $(3, 2)$ | 6.5 | 2 | 153 | 153/4 |
| $6, 8, 10$ | 2 | $(2, 1)$ | 5 | 2 | 84 | 84/4 |

Here $R = \frac{k}{2}(m^2 + n^2)$ and $r = kn(m - n)$, and since $c = 2R$ the area factorises as $N = 4(R - r)(R + r) = uv$ with $u = 2(R - r) = c - 2r$ and $v = 2(R + r) = c + 2r$, which at $k = 1$ are $(m - n)^2 + 2n^2$ and $(m + n)^2 - 2n^2$, so $N = k^2uv$ $(84 = 4 \cdot 21)$.

## 3 / READING DEGENERACY FROM $N$

### Equal annulus areas are factor pairs $(u, v)$ of the same $N$

Given $N$, a triangle with that annulus value has $c - 2r = u$ and $c + 2r = v$ for some $N = uv$, so $c = \frac{u+v}{2}$, $a + b = v$ and the leg difference $d = b - a$ satisfies $d^2 = \frac{u^2+2uv-v^2}{2}$:

$$uv = N, \quad v \equiv u \pmod 4, \quad \tfrac{1}{2}(u^2 + 2uv - v^2) = d^2$$

Each passing pair gives the triangle with legs $\frac{v \mp d}{2}$, so the degeneracy of $N$ is the number of passing pairs; for $(9, 17)$, $d^2 = \frac{81+306-289}{2} = 49$ and the legs are $5, 12$. For primitive triples $u = x^2 + 2y^2$ and $v = X^2 - 2Y^2$, so a prime that moves from $u$ to $v$ between two triangles must be $1 \pmod 8$.

## 4 / THE FIRST COINCIDENCES

### Different triangles with exactly the same annulus area

| $N$ | kind | the two triangles |
|---|---|---|
| 12 807 081 | enlarged and primitive | $21 \times (104, 153, 185) = (2184, 3213, 3885)$ from $(13, 4)$<br>$(2573, 2964, 3925)$ from $(57, 26)$ |
| 112 480 741 | enlarged and primitive | $437 \times (7, 24, 25) = (3059, 10488, 10925)$ from $(4, 3)$<br>$(7579, 8820, 11629)$ from $(98, 45)$ |
| 87 110 147 844 489 | both primitive | $(6559293, 7845076, 10225925)$ from $(2897, 1354)$<br>$(3820324, 9044157, 9817925)$ from $(3071, 622)$ |

In $21 \times (104, 153, 185)$ the bracket is the primitive triple Euclid's formula gives for $(m, n) = (13, 4)$ and 21 enlarges it, so $(m, n)$ is an input and not a triple. The first two coincidences pit an enlarged triangle against a primitive one. Between two primitive triples there is none up to hypotenuse $10^7$, and exactly two pairs up to $10^8$, the first having $N = 3 \cdot 7 \cdot 11 \cdot 401 \cdot 449 \cdot 457 \cdot 4583$.

## 5 / HOW COMMON IS IT

### Rare: none below hypotenuse $3925$, and no value shared by three triangles

| hypotenuse | integer right triangles | values of $N$ shared | new pairs (not rescalings) |
|---|---|---|---|
| $\le 10\,000$ | 12 471 | 2 | 1 |
| $\le 100\,000$ | 161 436 | 43 | 9 |
| $\le 400\,000$ | 733 932 | 197 | 16 |

## ASIDE / THE TANGENT CHORD

Equal area means equal tangent chord $L = \sqrt{N}$ (3578.7 for the first pair).

## LINKS TO OTHER SHEETS

Radii: the incircle sheet; parameters: the triples sheets; $\mathbb{Z}[\sqrt{2}]$ norms: the Pell sheet.

**Convention.** $N = 4A/\pi = c^2 - 4r^2$; $k \times (a, b, c)$ is a primitive triple enlarged by $k$; counts are exact searches.

Constructed by Silicon · Anthropic Claude — Architect Carbon · LeeMcCullochJames